



Patterns of Specialty Tobacco Retail Locations and Visitor Counts in the United States

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Abstract

Background: Specialty tobacco retailers, including vape shops, smoke shops, and cigar shops, are important sources of tobacco access, yet there is no reproducible, national database documenting their locations and consumer engagement.

Methods: We develop a national database of specialty tobacco retailers in the United States by combining Google Maps business listings as of June 2024 with Advan unique visitor count data, updated to May 2024. We match approximately 20,000 businesses across both databases and observe roughly 50 million monthly visits to these locations. We then perform two cross-sectional analyses examining specialty retailer density and visits at the census tract and county levels.

Results: We find strong clustering of all types of specialty retailers in low-income areas and fewer shops per capita in rural areas. We find that smoke and vape shop density is negatively correlated with both educational attainment and population share of racial and ethnic minorities. Cigar and smoke shops were less common in areas with high shares of young people ages 15–24. Additional businesses are positively associated with total consumer visits, with evidence of spillover effects across shop types.

Conclusions: This study provides the first reproducible, national assessment of specialty tobacco retailer locations linked to consumer visit data, offering a new foundation for monitoring retail access and informing tobacco control policy.

Introduction

Specialty tobacco retailers are a little understood aspect of the tobacco retail supply side. In this manuscript, we identify and analyze three categories of specialty tobacco retailers: vape shops, smoke shops, and cigar shops. These retailers have different customer bases and may make location decisions to maximize accessibility among their clientele.^{1,2}

There is no publicly available national registry of all tobacco retailers. Specialty tobacco retailers are not well captured by popular point-of-sale scanner systems like Nielsen and Information Resources Incorporated (IRI, now Circana). As a result, less is known about

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specialty tobacco retailers than about larger and more conventional retail outlets. Liber et al.³ estimate that scanner-based sales systems account for a substantially larger share of cigarette sales than electronic nicotine delivery systems (ENDS) sales, highlighting the potential importance of specialty retailers and online retailers for understanding the ENDS market. We contribute to literature identifying tobacco retailer locations by combining Google Maps data with Advan visitor patterns to catalog specialty retailer locations not identifiable in point of sale scanner data. We then analyze associations between retailer locations, area-level sociodemographic characteristics, and visitor counts to these locations.

Tobacco product access is central to tobacco control policy. Neighborhoods with higher concentrations of tobacco retailers have been found to be positively associated with tobacco use. Policies restricting access—including Minimum Legal Sales Age laws such as Tobacco-21 and restrictions on flavored tobacco products—have also been shown to affect tobacco purchasing patterns and use⁴⁻⁶. In some cases, policy effects may extend beyond their immediate jurisdictions, as consumers may travel across borders to purchase products restricted in their home jurisdictions.⁷

Prior research has relied on the number of retailers as a measure of access. For example, counts of tobacco product retail locations increased by 12% in the United States from 2010-2017.⁸ However, this approach treats all retailers as equal, overlooking substantial variation in size, popularity, and sales volume. A single high-volume retailer may facilitate as much access as several smaller outlets. Since direct measures of establishment volume (e.g., square footage, inventory, or sales) are not systematically available for specialty tobacco retailers, we use visitor counts as measured by cell phone tracking data as a proxy for retail exposure and potential consumer access. Prior research has found foot traffic and retail sales to be highly correlated in other contexts and used in other public health literature to proxy for consumption.^{9,10} This measure captures not only the presence of a retailer, but also how actively consumers engage with it, offering a behaviorally relevant indicator of access.

Studies of specialty tobacco retailers have also mostly focused on limited geographies such as selected counties or Metropolitan Statistical Areas.^{2,11-14} At the national level, some previous literature has developed methods for identifying likely retailers of certain products such as premium cigars in the National Establishment Time Series (NETS) data.¹⁵ Other attempts have relied on the use of online directories such as Yelp, YellowPages, and [GuidetoVaping.com](https://www.guidetovaping.com) to catalog locations nationally,^{1,16} in addition to Google.¹⁷

This study builds on previous research on specialty tobacco retailer locations in several ways. First, it creates a nationwide database of specialty tobacco retailers and distinguishes between three main store types: vape shops, smoke shops, and cigar shops. Second, our registry is very recent, as of June 2024, thus providing the first national estimate since 2018 for vape shops¹⁶ and for the first time ever for smoke and cigar shops. Third, our registry is linked to visitor count data as recorded by Advan, which uses cell phone tracking data across over 40 million devices.¹⁸ To demonstrate the utility of this novel database, we then performed two, regression-based analyses to examine area-level correlates of shop density and the relationship between shop density and visit patterns.

Data and Methods

To create a database of specialty tobacco retailers publicly visible in Google Maps, we used a third party program to iteratively query Google Maps in June 2024. We use Google Maps rather than alternative business databases because prior work by Pearson et al.¹⁹ found that, among several online sources reviewed, Google listings had the lowest degree of business-type misclassification. For each ZIP code, we searched Google Maps for businesses whose primary store type matched one of the following categories: vaporizer store, vaporizer shop, tobacco shop, tobacco supplier, smoke shop, or cigar shop. These specialty retail listings exclude larger retailers, categorized as department or grocery stores in Google, in addition to convenience stores and gas stations which may also sell tobacco products.

Because Google listings can include multiple categories, we used only the primary business type to assign mutually exclusive shop categories and avoid double counting. These classifications are intended to capture each store's primary focus: vaporizer products, premium and loose tobacco products and accessories, or cigars. Although these categories may not perfectly reflect all products sold, they provide a reproducible approach for distinguishing vape, smoke, and cigar shops at national scale.

We found 45,315 unique business listings marked as operational with Google business ID and full address details. Since it was possible for businesses to appear in multiple queries based on secondary business type, reviews, or webpage keywords, we removed any duplicates by retaining only the single Google business ID at each address with the largest number of reviews. Finally, using the main business type categorizations visible on the Google Maps page, we filtered query results into mutually exclusive categories: vape shops (12,032 results), smoke shops (30,462 results), and cigar shops (2,821 results).

To provide additional insights into visitors at specialty tobacco retailers nationwide, we use Advan visitor count data as of May 2024.¹⁸ Advan aggregates Safegraph's cell phone mobility data, which tracks user movement to over 150 million points of interest (POI). At the time of analysis, Advan data included approximately 40 million cell phone users, equivalent to about 13% of the number of cell phone users in the U.S. Using a fuzzy matching process, we matched Advan to Google data using geographic coordinates, business names, and addresses from each database. Additional information on the data sources and matching procedure is included in the Online Data Appendix.

The matching process yielded 20,271 specialty retailer locations with available visitor count data actively operating in the United States, including 5,930 vape shops (49.2% match rate), 12,847 smoke shops (42.2% match rate), and 1,494 cigar shops (53.0% match rate). We retained both the matched and unmatched sets of retailers for subsequent analysis. The imperfect match rate likely reflects structural differences between the datasets—including differences in POI classification schemes, business naming conventions, spatial precision, and establishment coverage.

We then conducted two regression-based analyses. The first uses the full universe of retailers identified in the Google database to estimate the association between census tract-level demographics and shops per 100,000 tract residents. Census tracts are geographic entities

within counties and are defined to be as homogeneous as possible with respect to population characteristics, economic status, and living conditions.²⁰ We regress shop counts per capita on the following census tract-level characteristics as measured in the 2019 American Community Survey 5-year estimates data: percent of the population ages 15-24, percent female, percent racial or ethnic minority, percent below the federal poverty threshold, and percent with high school degree or less. We also control for whether the census tract was in a rural county as classified by the United States Department of Agriculture (USDA) Economic Research Service (ERS) rural urban continuum codes (RUCC). Following convention, we define rural counties as those with RUCC codes ≥ 4 . In our regression analyses, we weight by tract population size, include state fixed effects, and estimate Huber-White robust standard errors.

Using the subset of specialty tobacco retailers matched with visitor count data, we then characterize consumer engagement with specialty tobacco retailers by analyzing visitor counts to store locations as a function of store density at the US county level. We use counties as a reasonably small level of geography capturing resident tobacco purchasing. Additionally, prior validation work shows that sampled device counts are more reliable at the county level than at the tract or block-group levels.²¹ County aggregation also reduces sparsity, as most tracts contain no specialty stores, yielding a more interpretable estimate of the association between additional retailers and visitor volume across a broader local retail market.

In each specification, we regress the log number of visitors to a given shop type on the number of shops of that type, controlling for the number of other shop types, the same set of area-level characteristics (aggregated to the county level), and state fixed effects. This allows us to examine how an additional shop is associated with visits to that same type of shop – conditional on the number of other specialty shops – as well as any “spillover” effects on visits to other shop types. In each regression, we also control for the number of shops of each type that are unmatched in the Advan data. As in the prior analysis, we weight each regression by county population.

To demonstrate the structure of the dataset, we provide a limited supplemental sample with aggregate store counts, visitor counts, and match rates for a the largest 10% of states and counties by population. The file is provided for documentation purposes only; access to the licensed proprietary data requires appropriate data access rights from Dewey Data.

Results

To illustrate the geographic dispersion of specialty tobacco retailers in the US, we aggregate the number of each shop type to the county and census tract levels. Figure 1 displays the density deciles of all combined vape, smoke, and cigar shops, scaled per 100,000 residents in each county. Density patterns align with expectations in a few key places. For example, in Figure 1, Clark County, Nevada (home of Las Vegas) and Appalachia (with higher than average population smoking rates, see Horn et al.²²) have high density of specialty tobacco retailers; meanwhile, Utah (home to a sizable Mormon population) has low density. Figure 2 shows each shop type separately across three maps. Cigar shop locations are less common

nationally than vape and smoke shops, and they are disproportionately concentrated in Pennsylvania, Florida, and Washington DC. This aligns with expectations given that these are the only US jurisdictions that do not tax large cigars. ²³

Of the 72,257 census tracts in our data, spanning all 50 U.S. states and Washington, DC, 26,307 (36%) contained at least one specialty tobacco retailer as of June 2024. Among tracts with retailers, 58.6% have exactly one retailer, 24.3% have exactly two retailers, 9.9% have three retailers, and 7.2% have four or more. Table 1 presents summary statistics for the tracts included in our study. The population weighted average census tract had 0.20 vape shops (sd = 0.54), 0.49 smoke shops (sd = 0.88), and 0.045 cigar shops (sd = 0.23). On average, smoke shops received the most visitors, followed by vape shops and cigar shops.

Figure 3 presents regression coefficients and 95% confidence intervals showing the joint association between the density of each shop type and five census tract-level characteristics (age 15-24 %, racial/ethnic minority %, poverty %, rural (0/1), and no college degree %). The most salient of our findings is a large positive relationship between poverty and density of all shop types. In particular, a one percentage point increase in the tract poverty rate is associated with approximately 0.054 extra vape shops, 0.244 extra smoke shops, and 0.022 extra cigar shops (all per 100,000 people, $p < 0.01$).

A second consistent finding across all shop types is a negative correlation between shop density and rurality. Tracts in rural counties had on average 0.009 fewer vape shops, 0.017 fewer smoke shops, and 0.003 fewer cigar shops (all per 100,000 people, $p < 0.01$). Third, we also find statistically significant negative associations between the density of vape and smoke shops and the non-White population share. A one percentage point increase in the non-White population share is associated with 0.027 fewer vape shops and 0.017 fewer smoke shops (per 100,000 people, $p < 0.01$). In Appendix Figure 1, we disaggregate total minority share into share non-Hispanic Black, non-Hispanic Asian, non-Hispanic other, and Hispanic.

Some of our results were not consistent across shop types. For example, vape and smoke shop density is negatively correlated with educational attainment, while cigar shops demonstrates the opposite pattern. The density of smoke and cigar shops is negatively correlated with higher shares of young people (ages 15-24 years), but there is no statistically significant relationship with vape shop density.

Table 2 presents estimates of the association between shop density and visitor counts to each of the three specialty tobacco retailers at the county level. County-level visitor counts are log transformed, allowing coefficients to be interpreted as semi-elasticities representing the approximate percentage change in visitors associated with an additional shop located in the county, conditional on the number of matched and unmatched shop counts, county-level sociodemographic characteristics, and state fixed effects.

Results from this cross-sectional analysis suggest that an additional matched vape shop (representing a 53% increase from the mean) in the average county corresponds to 6.2% additional visits to vape shops, conditional on the number of other shops and model controls. We find that an additional vape shop is also associated with 3.8% more visits to smoke shops

and 7.6% more visits to cigar shops. An additional matched smoke shop (representing a 25% increase from the mean) is associated with 0.6% more visits to smoke shops, but 1.0% fewer visits to cigar shops. Finally, we find that an additional cigar shop (representing a 208% increase from the mean) is associated with approximately 17.7% additional visits to cigar shops and 3.4% fewer visits to smoke shops.

Discussion

This study provides a reproducible national assessment of specialty tobacco retailer locations and consumer visitor counts. Our study is unique in that it constructs and analyzes vape, smoke, and cigar shops individually and is the first to link specialty tobacco retail locations to visitor count patterns. Our results highlight heterogeneity in the association between specialty retailer locations and visitor counts, as well as with tract-level measures such as rurality, poverty rates, share of young people, racial composition, and education.

The total number of specialty tobacco retailers we identify in our Google dataset is larger than those found in prior literature. For example, we find 21% more vape shops ($n = 12,032$ shops) in 2024 than Dai et al.¹ in their examination of 2015 data from [Yellowpages.com](#) and [Guidetovaping.com](#) ($n = 9945$ shops). Similarly, we find 61% more vape shops than Venugopal et al.¹⁶ in their use of 2018 data from ReferenceUSA, Yelp, and Google ($n = 7,479$ shops). The larger number of vape shops we identify is likely due to our use of more recent data and increasing use of e-cigarettes over time among adults; between 2004–2022, the prevalence of current e-cigarette use among US adults increased from 3.7% to 6.0%, or a 62% increase.²⁴

Notably, we identify more cigar shops ($n = 2,821$ shops) than a recent National Academies of Sciences, Engineering and Medicine (NASEM) report, which documents 1,291 retailers registered by the Premium Cigar Association in 2021.²⁵ Our estimate of cigar shops – which is over twice that listed in the NASEM report – could be on account of many cigar specialty shops not being members of the Premium Cigar Association (thus avoiding membership dues).

Our results provide fresh evidence that specialty tobacco retailers cluster at the sub-county level in low income communities,^{26,27} where there are frequently higher rates of tobacco use.^{28,29} The positive association we observe between poverty and shop density across all shop types mirrors a recent systematic review, which reported significant positive links between socioeconomic disadvantage and tobacco retailer availability in 67 of 80 estimates.³⁰ This area-level correlation may contribute to income-based disparities in smoking prevalence, which have persisted despite overall reductions in smoking in the US.³¹

We found a significant negative association between vape shop density and minority share of the population. This adds to a mixed literature from state- or city-specific studies that have found evidence of both positive^{1,11} and negative^{32,33} associations between vape shop availability and racial/ethnic minority share at the census tract-level. Like vape shops, we also find that smoke shops are less likely to concentrate in census tracts with higher minority

shares. This stands in contrast with a recent systematic review of 41 studies in which 80% of study estimates identified positive associations between tobacco retailer density and racial/ethnic minority population share.³⁰ Our findings may diverge from prior work due to the recency of our data and the fact that many studies use broader definitions of tobacco-selling establishments (e.g., convenience stores, gas stations), which may differ from our focus on specialty tobacco retailers. A further distinction is that our models include state fixed effects, so estimates are identified from within-state variation and do not reflect between-state differences in policy environments, retail markets, or anti-smoking sentiment.

We also found that specialty retailers are less likely to operate in rural areas, where smoking prevalence is often higher,³⁴ but vape use has historically been lower.³⁵ One possible explanation is that people in rural communities may be more likely to use non-specialty retailers, like grocery stores and gas stations, to purchase tobacco. Our finding aligns with broader retail patterns, as other niche store types such as specialty grocers are also rarer in rural areas.³⁶

Finally, smoke and cigar shops appeared to cluster in census tracts with lower shares of 15–24 year olds, while there was a positive but statistically insignificant relationship between youth share and vape store density. This could be due to zoning laws that restrict tobacco retailer presence within certain distances of schools.³⁷ It could also reflect lower demand for tobacco among young people in general as tobacco use rates, including e-cigarettes, have fallen 60% among young people since 2011.³⁸

One unique contribution of our paper is that we test whether additional specialty tobacco retailers in a county are associated with higher visitor counts on net, or simply redistribute existing visitors across multiple shop types. Our results suggest that an additional vape shop is associated with more visitors to all three types of specialty tobacco retailers. However, given the cross-sectional nature of our data, we cannot determine whether vape shops increase visits to other shop types or locate in areas with already high demand. In future work, we will harvest Google data annually and link repeated cross-sections to Advan visits, creating a panel of specialty tobacco retailer supply and consumer visits. Such data will better support natural experiment designs to estimate the causal effects of tobacco retail policies.³⁹

A second limitation is that our classification of shop types does not perfectly predict product offerings, potentially leading to misclassification. For example, prior work has found that vape shop locations drawn from Yelp had a high probability of selling tobacco products in addition to e-cigarettes.⁴⁰ Moreover, using the primary business type only may overstate the number of retailers selling nicotine-containing tobacco or vaping products if some included businesses primarily sell glassware, CBD products, or other non-nicotine products. Although Google listings incorporate owner-submitted information, user feedback, reviews, photos, and web-based sources, classification error likely remains. While manual verification of all retailers was not feasible for a nationwide database, future work could validate a stratified sample using manual review, licensing lists, or third-party commercial data.

Third, our study is specifically of vape, smoke, and cigar shops, and so we make no effort to study other retailers that sell tobacco as part of broader product offerings, such as grocery stores and supermarkets, corner stores, gas stations, and convenience stores. Prior work has found that non-specialty stores account for approximately 80% of all tobacco retailers in the US, ⁸ suggesting that we capture a small share of the tobacco marketplace. However, our focus on these retailers is important as they are not available through standard retail scanner database systems, thus requiring alternative methods of surveillance such as those we employ in our study. Further, these stores are frequently the subject of specific policy interventions such as retailer licensing and zoning restrictions.

Fourth, we are unable to verify the precise frequency with which Advan updates its point-of-interest dataset. Although we used the most current visitor count data available as of May 2024, prior large-scale POI additions in 2019 and 2023 suggest that Advan updates may lag Google Maps, which allows businesses to update listing information continuously. We are therefore more likely to match on store locations which have existed longer as they are most likely to have been updated into both databases. Future work incorporating panel data analysis can limit this concern by specifically identifying new versus preexisting locations in the data.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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What is already known on this topic

Prior studies have examined tobacco retailer density in the United States, focusing on limited geographic areas such as US states or metropolitan areas, or specific products such as e-cigarettes or cigars.

What this study adds

We provide the first U.S. national registry of specialty tobacco retailers linked to visitor count data, distinguishing among vape, smoke, and cigar shops. Our findings show that specialty retailers cluster in low-income areas, rural tracts have fewer shops despite higher smoking prevalence, and that additional vape shops increase overall visits.

How this study might affect research, practice or policy

This database enables monitoring of consumer engagement with specialty tobacco retailers, which are understudied sources of tobacco access. By linking shop locations to consumer visits, the study provides a new evidence base for evaluating retail regulations, enforcement, and equity impacts of policies restricting tobacco product availability.

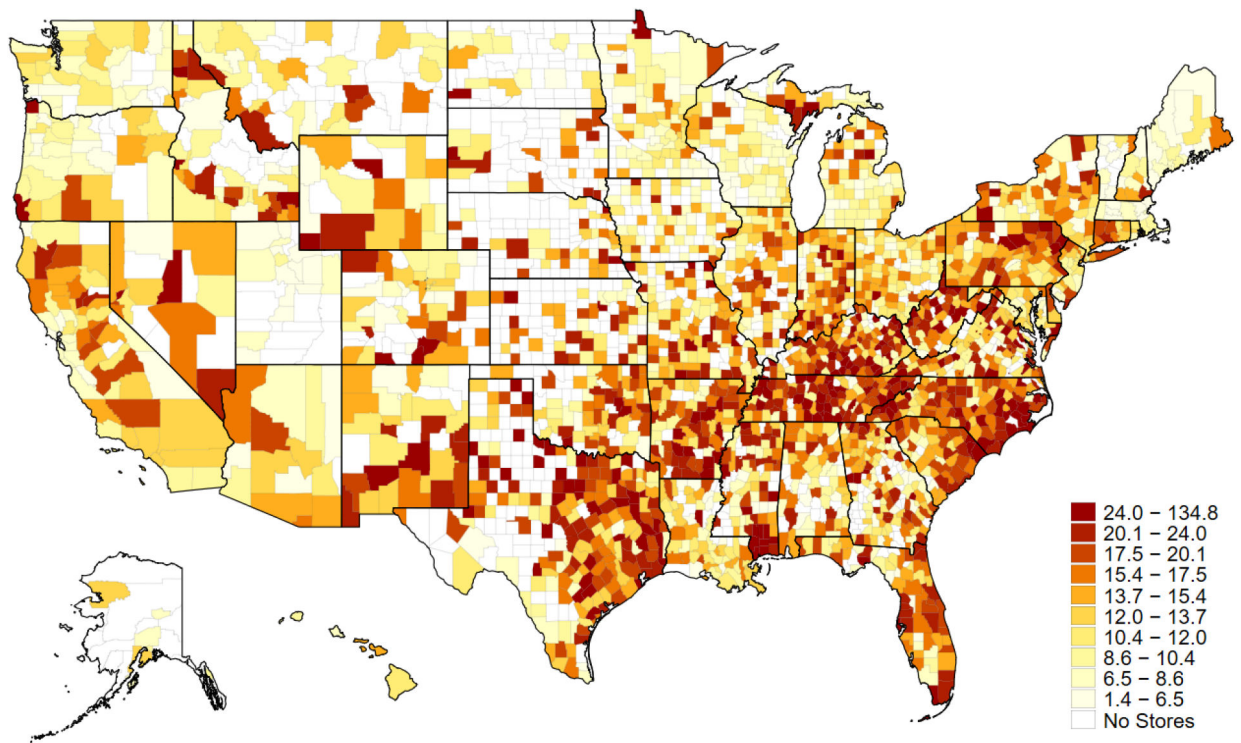
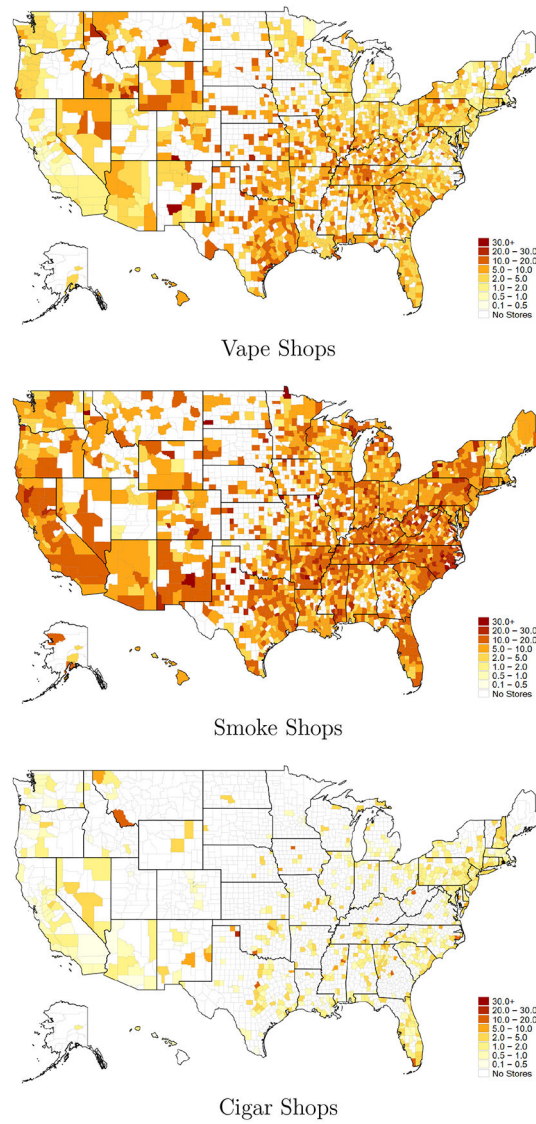


Figure 1:

Density of Specialty Tobacco Retailers, per 100k Population

Note: Figure 1 displays deciles of the number of specialty tobacco retailers (combined vape, smoke, and cigar shops) per 100k county residents, based on Google data.

**Figure 2:**

Density of Shop Types by County, per 100k Population

Note: Figure 2 displays deciles of the number of vape, smoke, and cigar shops per 100k county residents, based on Google data.

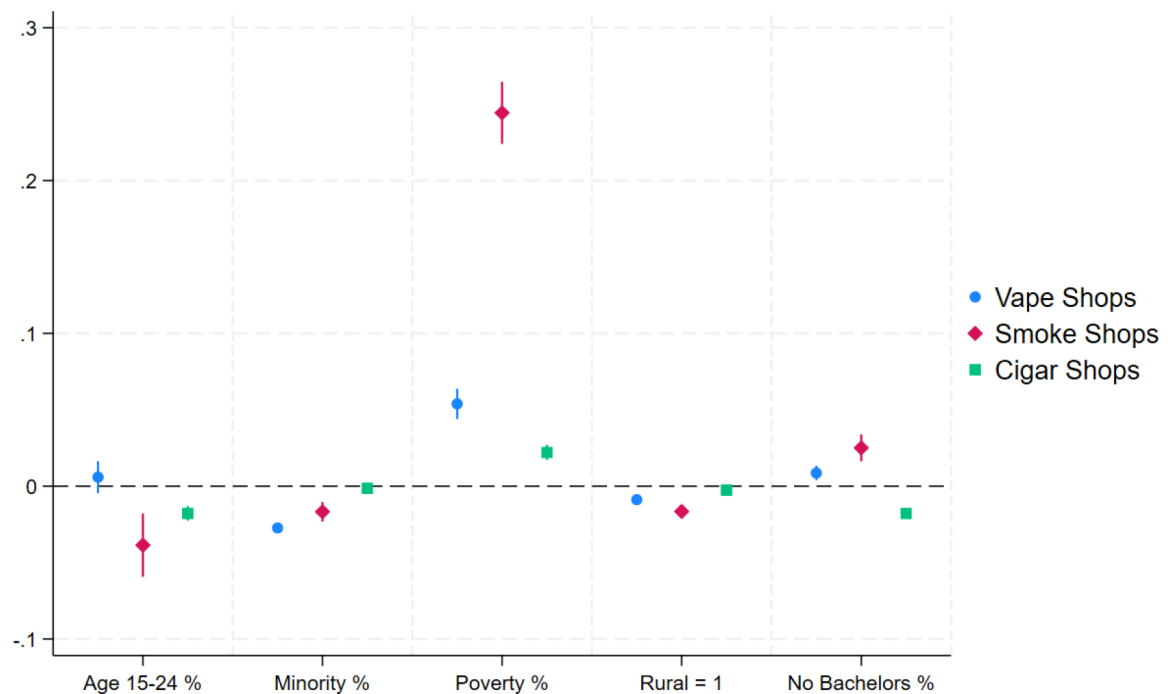


Figure 3:

The Association of Census Tract Demographics with Density of Vape, Smoke, and Cigar Shops

Note: N= 72,257 tracts. Coefficients are derived from an OLS census tract level regression with population weights, jointly controlling for each of the five demographic variables listed on the graph, female %, and state fixed effects, and with robust standard errors. Intervals are drawn to show 95% confidence bands.

Table 1:
Population Weighted Census Tract Summary Statistics

	mean	sd	min	max
Vape Shops	0.2025	0.5360	0	17
Smoke Shops	0.4879	0.8803	0	29
Cigar Shops	0.0452	0.2291	0	7
Population	4444.5160	2349.0230	0	72041
Age 15-24 %	13.3119	8.1059	0	100
Minority %	39.2973	29.3938	0	100
Female %	50.7582	4.2771	0	100
Poverty %	13.6148	10.7827	0	100
No Bachelors%	68.2859	19.0439	0	100
Rural tract %	16.8338	37.4168	0	100
Vape Visitors	20.5673	149.3086	0	12579
Smoke Visitors	40.1826	203.9492	0	15236
Cigar Visitors	6.8056	249.1932	0	60205
N	73,071			

Note: Table 1 presents summary statistics for all variables included in the analysis presented in Figures 1-3. Tract shop counts reflect the full Google database; visitor counts reflect matches between Google and Advan.

Table 2:
Association between Specialty Tobacco Retailer Density and Visitor Counts, by Shop Type

Logged Visitors in All Counties			
	ln(Vape Visitors)	ln(Smoke Visitors)	ln(Cigar Visitors)
# Vape Matches Avg = 1.89	0.062 *** (0.009)	0.038 *** (0.008)	0.076 *** (0.014)
# Smoke Matches Avg = 4.08	0.001 (0.003)	0.006 ** (0.002)	-0.010 * (0.005)
#Cigar Matches Avg = 0.48	-0.016 (0.025)	-0.034 * (0.018)	0.177 *** (0.048)

Note: N=3,141 counties. Outcomes are specified as log(1+visitors) to accommodate zero-visit observations. Each column is derived from a separate OLS county-level regression with population weights, jointly controlling for matched store types (displayed), unmatched store types, age 15-24 %, minority %, poverty %, rurality, less than bachelor %, female %, and state fixed effects. Robust standard errors are in parentheses.

*
 $p < .1$

**
 $p < .05$

 $p < .01$